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MINISTRY OF
POWER



India Energy Stack (IES)



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IES is for Power what UPI is for Banking

In 2014, India had banks, phones, and money — but no common digital rails. Then UPI happened. Not a new bank. Not a new app. Just a protocol. It unleashed a \$2 trillion digital payments economy.

**INDIA'S POWER SECTOR TODAY LOOKS EXACTLY
LIKE BANKING DID IN 2014**



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We Have the Pieces Now We Build the Rails



Rooftop Solar Owner, Jaipur

Generates surplus power every afternoon. Can't sell it to her neighbor on a different DISCOM.



EV Charging Startup, Bengaluru

Has to build a separate integration for every charge point operator in every city.



Energy Regulator, Chennai

Receives the same data in different Excel formats from different utilities.



Dairy Cooperative, Rural Maharashtra

Could save 40% on electricity by shifting cold storage off-peak. No one can measure or pay her for it.



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IES Enables Coordination at Scale

India has made significant investments in

SCADA systems - 200+

smart meters - 52.8 Million (as of Dec'25)

Interfaces - 200+

WITH IES

- From Silos to Shared Information
- Interoperability by Design
- Open Innovation Ecosystem



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What If Energy Had a Trust Bridge?

Every meter, solar panel, and EV charger has a verified digital identity

Like Aadhaar — but for energy assets

Any app can discover, transact, and settle energy services across any utility

Like UPI — but for electrons

A regulator writes a tariff rule once, every billing system executes it identically

Like GST Network — but for power

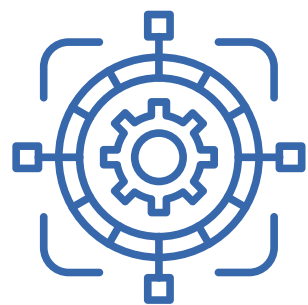
A prosumer sells surplus solar to any buyer, verified and settled automatically

Like ONDC — but for energy markets



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This is the India Energy Stack



A **set of protocols or specifications** enabling **uniform, reliable, and trustworthy interoperability** between grid entities and consumers.



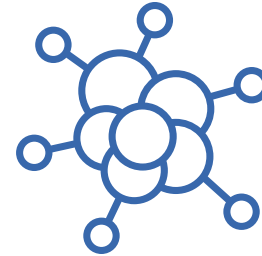
A set of services (API definitions and calls) and a **taxonomy/architecture** describing how components interact.

IES is to energy what HTTP is to the internet — **it doesn't build the websites, it lets them talk to each other.**

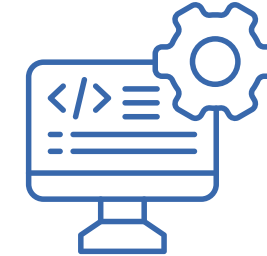
IES IS NOT



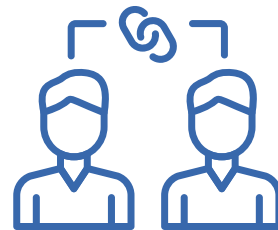
A centralised
database or **data lake**



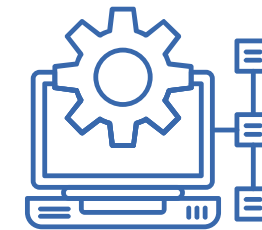
A centralised service
pulling data from entities



A software **product**



Dependent on strict hierarchy in the
energy sector



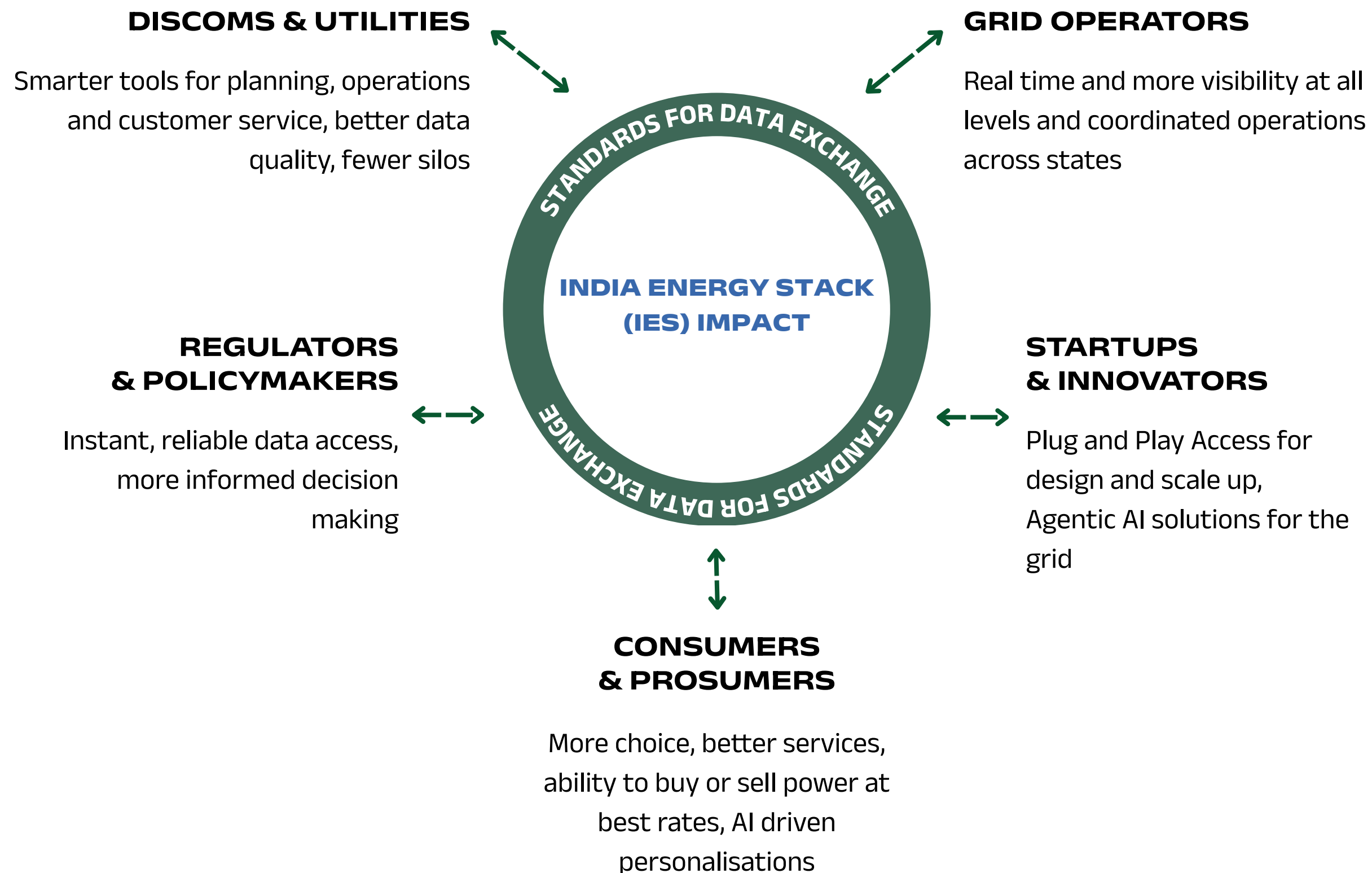
A tool to **integrate internal systems**
of any stakeholder



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Impact on stakeholder



IES WILL BE ROLLED OUT THROUGH THREE KEY INITIATIVES:



IES Architecture

- Defines the overall IES ecosystem architecture, including the blueprint, core building blocks, protocols, and supporting specifications.
- Establishes the framework that enables transactions and data exchange across all ecosystem actors.



IES Adoption Strategy

- Identifies programs and policy initiatives to incentivise and encourage adoption across the IES ecosystem.
- Targets relevant entities and stakeholders to drive ecosystem-wide participation.



IES Accelerator

- Implements sandbox environments, tools, and reference solutions to support ecosystem readiness.
- Provides key enablers needed to accelerate IES adoption across all stakeholders.

Built on foundational building blocks



Identity & Addressing

Unique IDs for every consumer, meter, and grid asset



Registries & Directories

Who is authorized and where to find their services



Interaction Protocols

Standard rules for exchanging data across systems



Energy Credentials

Verifiable digital proofs of eligibility and authority



Energy Data Exchange

Tariffs and rules as machine-readable, executable logic and well as digital data reporting



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IES Use Cases- informing design

Issuance of Energy Credentials

One Digital Record -
for PoA, schemes
and participation



Peer to Peer Energy Exchange

Send power from one
premise to another



DER Integration

Every Solar Panel, Battery, and EV is
Visible and Manageable to offer flexibility



Public Energy Data Exchange

Make policies
machine readable



Regulatory Data Exchange

Scattered Reports
to Real-Time
Intelligence





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From Energy Access to Energy Agency

**IES creates a new
class of energy
micro-entrepreneurs
enabling livelihoods**

Self-Help Groups in Odisha

Enable communities to co-own DER assets, trade surplus power, and energise local enterprises

College Student in Pune

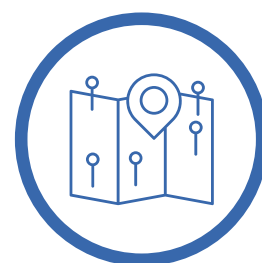
Builds an energy app in a hackathon. It works with every DISCOM in India on day one.

Farmer in Madhya Pradesh

Earns by shifting pump timing to off-peak hours. The system measures, verifies, and pays — automatically.

Consumers become participants. Passive becomes active. Access becomes agency.

IES SUCCESS METRICS & KPIs



Adoption and Coverage

- % of utilities aligned with IES specs
- % of interactions through IES interfaces



Ecosystem Development

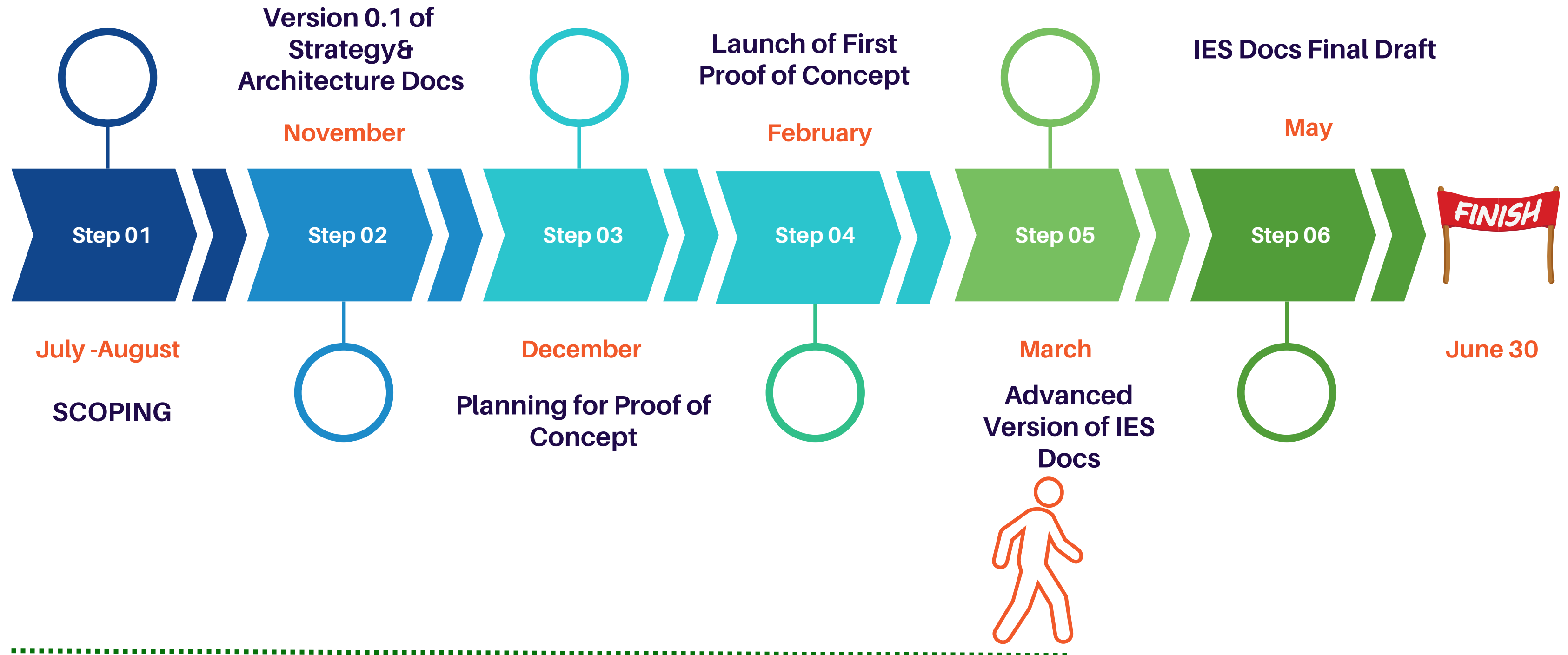
- Number of 3rd-party apps leveraging IES
- Growth in sandbox participants



Consumer Value

- Improved consumer access and choice
- Reduction in transaction costs

PROJECT KEY MILESTONES





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One Nation, One Frequency, and now One Digital Grid

BUILT IN INDIA, FOR THE WORLD

THANK YOU

CONTACT

Nodal Agency: IES@recindia.com

Knowledge Partner: IES.Secretariat@fsrglobal.org

